

DSA Lab 9 Practice Problems

Topic: Minimum Spanning Trees and Topological Sorting

Resources

Minimum Spanning Trees

Note: The course covers two algorithms for finding a minimum spanning tree of a graph, one of which relies on the use of a so-called *disjoint-set data structure*, so you will need to learn about said data structures first.

CLRS (the course textbook) (recommended):

- Chapter 19 (data structures for disjoint-sets)
- Chapter 21 (minimum spanning trees)

[Striver's graphs playlist](#) (videos 44-53)

[VisuAlgo](#)

Topological Sorting

CLRS: Section 20.4

[Striver's graphs playlist](#) (videos 21-26)

Minimum Spanning Trees

Easy

[Number of Provinces](#)

[Road Construction](#)

[Min Cost to Connect All Points](#)

Medium

[Redundant Connection](#)

[Number of Operations to Make Network Connected](#)

[Most Stones Removed with Same Row or Column](#)

[Accounts Merge](#)

[Path with Minimum Effort](#)

[Make It Connected](#)

[Shichikuji and Power Grid](#)

Hard

[Swim in Rising Water](#)

[Find Critical and Pseudo-Critical Edges in Minimum Spanning Tree](#)

Topological Sorting

Easy

[Topological Sort](#)

[Course Schedule](#)

[Course Schedule II](#)

Medium

[Find Eventual Safe States](#)

[Find All Possible Recipes from Given Supplies](#)

[Alien Dictionary](#)

[Parallel Courses](#)

Hard

[Build a Matrix With Conditions](#)

[Sort Items by Groups Respecting Dependencies](#)

Maximum Employees to Be Invited to a Meeting